Linear Transformations and Their Matrices

Linear Algebra - CS 3A and 3B

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Example 11: $\mathbb{R}^4 \to \mathbb{R}^2$ — Linear functional

Problem: $T(x_1, x_2, x_3, x_4) = (x_1 - x_3, 2x_2 + x_4)$. Find matrix.

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Apply to $(2,3,4,5)$:

$$T = \begin{pmatrix} 2-4 \\ 2\cdot 3 + 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 11 \end{pmatrix}$$

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Note: This is an **idempotent** transformation: $A^2 = A$.

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$$\text{proj}_{\mathbf{n}}\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{n}}{\|\mathbf{n}\|^2}\mathbf{n} = \frac{x + y + z}{3}(1, 1, 1)$$

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Apply to $(1, 0, 0)$: $\frac{1}{3}(2, -1, -1)$ which lies in $x + y + z = 0$

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Composition $T(\mathbf{x}) = A_{\text{refl}}(A_{\text{rot}}\mathbf{x}) = (A_{\text{refl}}A_{\text{rot}})\mathbf{x}$

$$A_{\text{total}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

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That's a reflection across x-axis! Check on $(1, 0)$: rotate$\to(0, 1)$, reflect$\to(1, 0)$; direct: $(1, 0)\to(1, 0)$

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Apply to $(1, 2, 3)$:

$$T = (1 + 6,\ 2,\ 3) = (7, 2, 3)$$

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Example: $T(x, y) = (x, 0)$ in $\mathbb{R}^2$.

$\ker(T) = \{(0, y)\} \Rightarrow \dim = 1, \quad \text{Im}(T) = \{(x, 0)\} \Rightarrow \dim = 1,\ 1 + 1 = 2$

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$$\ker(T) = \text{span}\{(1, -1, 1)\}, \quad \dim = 1$$

Check: $T(1, -1, 1) = (0, 0, 0)$

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Columns: $C_1 = (1, 0, 1), C_2 = (1, 1, 0), C_3 = (0, 1, -1)$. Check independence: $C_3 = C_2 - C_1$? $(0, 1, -1) = (1, 1, 0) - (1, 0, 1) = (0, 1, -1)$
So $\dim(\mathrm{Im}) = 2$, basis $\{(1, 0, 1), (1, 1, 0)\}$.

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So $\dim(\mathrm{Im}) = 2$, basis $\{(1, 0, 1), (1, 1, 0)\}$. Check dimension theorem: $\dim\ker = 1$, $\dim\mathrm{Im} = 2$, $1 + 2 = 3$

Example 18: $\mathbb{R}^4 \to \mathbb{R}^4$ — Rotation in 4D?

Problem: Rotation in $\mathbb{R}^4$ can be block diagonal:

$$A = \begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & \cos\phi & -\sin\phi \\ 0 & 0 & \sin\phi & \cos\phi \end{pmatrix}$$

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For $\theta = 90^\circ$, $\phi = 0$: $A = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ Apply to $(1, 0, 2, 3)$:

$$T = (0, 1, 2, 3)$$

First two coordinates rotate, last two unchanged.

Example 19: Differentiation as Linear Transformation

Problem: $T : P_3 \rightarrow P_2$ (polynomials degree 3 to 2) defined by $T(p) = p'$. Basis $\{1, x, x^2, x^3\}$.

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Matrix (using basis $\{1, x, x^2\}$ for codomain):

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Apply to $p(x) = 2 + 3x + 4x^2 + 5x^3$: coefficients $(2, 3, 4, 5)^T$.

$$A \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \\ 15 \end{pmatrix} \Rightarrow 3 + 8x + 15x^2$$

Check derivative: $p'(x) = 3 + 8x + 15x^2$

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Matrix (basis $\{1, x, x^2, x^3\}$):

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{pmatrix} \quad (4 \times 3)$$

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Apply to $p(x) = 4 + 5x + 6x^2$: $(4, 5, 6)^T$

$$T = \begin{pmatrix} 0 \\ 4 \\ 5/2 \\ 2 \end{pmatrix} \Rightarrow 0 + 4x + \frac{5}{2}x^2 + 2x^3$$

Check: $\int_0^x (4 + 5t + 6t^2)dt = 4x + \frac{5}{2}x^2 + 2x^3$

Summary Table of Transformations

Type	Matrix (2D)	Effect
Identity	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	No change
Rotation θ	$\begin{pmatrix} c & -s \\ s & c \end{pmatrix}$	Rotate by θ
Scale a, b	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	Stretch/compress
Shear x	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	Horizontal skew
Reflect x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Flip vertically
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Key: $c = \cos\theta$, $s = \sin\theta$; $k =$ shear factor.

Matrix of Transformation: Change of Basis

If B and C are bases for domain and codomain:

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$T(1, 1) = (3, 0) \to [3, 0]^T$, $T(1, -1) = (1, 2) \to [1, 2]^T$.

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If $\mathbf{v} = 2(1, 1) + 1(1, -1) = (3, 1)$: $T(\mathbf{v}) = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$. Direct: $T(3, 1) = (7, 2)$

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$$A^{-1} = -\tfrac{1}{2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$$

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$T^{-1}(u, v) = (-2u + v,\ 1.5u - 0.5v)$. Check: $T(1, 0) = (1, 3)$,
$T^{-1}(1, 3) = (-2 + 3, 1.5 - 1.5) = (1, 0)$

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Eigenvalues: $\lambda = 3$ (eigenvector $(1, 0)$), $\lambda = 1$ (eigenvector $(-1, 2)$).

Eigenvalues and Eigenvectors of Transformations

For $T : V \to V$, λ is eigenvalue if $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v} \neq 0$.
Example: $T(x, y) = (3x + y, y)$ from earlier.

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$$

Eigenvalues: $\lambda = 3$ (eigenvector $(1, 0)$), $\lambda = 1$ (eigenvector $(-1, 2)$).
Check $T(-1, 2) = (-3 + 2, 2) = (-1, 2) = 1 \cdot (-1, 2)$

Geometric Transformations in Computer Graphics

Common 3D transformations (homogeneous coordinates):

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & t_x \\ a_{21} & a_{22} & a_{23} & t_y \\ a_{31} & a_{32} & a_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

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Example: Translate $(1, 2, 3)$ by $(4, 5, 6)$:

$$\begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \\ 1 \end{pmatrix}$$

Summary and Key Insights

Every linear transformation:

- Is represented by a matrix
- Satisfies $T(c\mathbf{u} + \mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v})$
- Has kernel and image
- Composition = matrix multiplication

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Linear transformations are the language of data science, graphics, and machine learning!

Final Practice Problem

Find the matrix of $T : \mathbb{R}^3 \to \mathbb{R}^3$ that:

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$A = QR = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$ Check on $(0, 1, 0)$: reflect $\to (0, -1, 0)$, rotate $\to (0, 0, -1)$. Direct: $A(0, 1, 0) = (0, 0, -1)$

End — Thank You!